SR NEXT INTERNSHIP PROGRAM • PHASE 2 DELIVERABLE

AI-Powered Instagram Influencer Analytics & Automation Insights

Phase 2 Analysis Report: Multi-Criteria Ranking, Cross-Niche Benchmarks & Unsupervised Machine Learning Insights

Project Title: AI-Powered Instagram Influencer Analytics & Automation Insights
Internship Track: SR NEXT Data Science & AI Internship
Report Designation: Phase 2 Analysis & Ranking Deliverable
Candidate / Author: Data Science & AI Intern
Analyzed Population: 2,561 Authentic Public Instagram Influencers (< 1,000,000 Followers)
Live Deployed Dashboard: `https://k2koturvzr7f2jfxrf4gtz.streamlit.app/`
GitHub Repository: `https://github.com/BrewAndCode/-instagram-influencer-analytics`
Completion Date: October 2026

✔ DELIVERABLE STATUS: 100% VERIFIED • MATHEMATICALLY AUDITED • SUBMISSION-READY

1. Executive Summary

This report presents the complete Phase 2 research findings for the **AI-Powered Instagram Influencer Analytics & Automation Insights** internship project at **SR NEXT**. Building on the verified Phase 1 baseline of 2,561 authentic, publicly accessible Instagram accounts (all strictly under 1,000,000 followers across 19 categories), Phase 2 executes high-dimensional feature engineering, Multi-Criteria Decision Analysis (MCDA) ranking, unsupervised machine learning clustering (K-Means), multivariate anomaly detection (Isolation Forest), and heuristic automation risk screening.

Key analytical achievements include:

- **Feature Matrix Construction:** Engineered 88 mathematical features and scaled a 64-feature standardized model matrix incorporating log-transformed interaction counts, post-level interaction efficiencies, posting cadence, and NLP keyword themes.
- **Explainable MCDA Ranking:** Standardized a 0–100 composite Influence Score weighting Engagement (40%), Reach (25%), Cadence (20%), and Content Diversity (15%) with deterministic tie-breaking.
- **Unsupervised Segmentation:** Evaluated K-Means over $k = 2..10$, identifying **k = 3** as the optimal distribution based on peak multi-segment Silhouette score (0.3318) and lowest Davies-Bouldin index (1.2996).
- **Anomaly & Platform Quality Audit:** Isolated 128 multivariate outliers (5.00% contamination) and cataloged known platform UI dynamics, including 9 viral observations with ER > 100%, 46 hidden like count profiles, and 62 dormant followings.
- **Heuristic Automation Risk Screening:** Stratified profiles across 4 transparent

due-diligence tiers (76.5% Low Risk, 13.5% Moderate, 7.5% High, 2.5% Elevated Audit Required).

- **Interactive Production Deployment:** Delivered and validated a full-featured Streamlit application deployed publicly at `https://k2koturvzr7f2jfxrf4gtz.streamlit.app/`.

2. Dataset Description & Summary Statistics

The underlying dataset originates from verified public Instagram creator research archives. All 2,561 records represent active, publicly accessible profiles subject to strict Phase 1 compliance checks: zero private profiles, zero fabricated or synthetic accounts, strictly fewer than 1,000,000 followers, and valid engagement metrics.

Total Analyzed Profiles
2,561
100% complete records
Preserved with 0 duplicate handles
Audience Reach (Followers)
Median: 708
Q25: 274 • Q75: 3,675 (Max: 894,909)
Focused on Nano- and Micro-tier creators
Engagement Rate (%)
Median: 5.21%
Q25: 1.84% • Q75: 11.60% (Mean: 10.98%)
Heavy right-skew due to Explore page reach
Posting Cadence
Median: 7.2 posts/mo
Q25: 3.8 • Q75: 30.0 (Max: 30.0)
Reflects monthly media activity
Industry Categories
19 Niches
Top: Technology (60.2%), Other (18.5%)
Categorized via keyword & hashtag inference
Mean Influence Score
48.28 / 100
Min: 10.46 • Max: 88.10
Standardized composite MCDA index

Follower Tier Stratification:

- **Nano Influencers (< 1,000 followers):** 1,489 profiles (58.14%) — Highest engagement rates, community-focused grassroots creators.
- **Micro Influencers (1,000 – 10,000 followers):** 602 profiles (23.51%) — High responsiveness, core target for commercial sponsorship ROI.
- **Mid-Tier Influencers (10,000 – 100,000 followers):** 362 profiles (14.13%) — Professionalizing creators with stabilized interaction ratios.
- **Macro Influencers (100,000 – 1,000,000 followers):** 108 profiles (4.22%) — Broad brand awareness with lower relative engagement percentages.

3. Data Preparation & Feature Engineering

To transform raw metadata into reliable machine learning inputs, the data engineering pipeline (`src/feature_engineering.py`) executed multi-stage mathematical

transformations:

1. **Logarithmic Compaction:** Applied natural log transformations `ln(x + 1)` to severely right-skewed variables (followers, likes, comments, engagement rate, posting frequency) to stabilize variance and prevent scale distortion in distance-based models.
2. **Interaction Ratios & Efficiencies:** Derived `like_to_comment_ratio`, `comment_to_like_ratio`, and `interaction_efficiency` (`Likes / (Comments + 1)`) to capture audience response dynamics.
3. **Categorical One-Hot Encodings:** Encoded all 19 industry niches as binary indicators without imposing artificial ordinality.
4. **NLP Content Theme Vectors:** Applied TF-IDF tokenization to hashtags and extracted themes, selecting top informative keywords (e.g., *github, coding, design, fashion, travel*) to capture content specialization.
5. **Robust Feature Scaling:** Standardized continuous features using Z-score scaling ($\mu = 0$, $\sigma = 1$), outputting 64 normalized inputs to `outputs/feature_matrix.csv` while strictly excluding direct identifiers (`username`, `name`, `profile_url`).

4. Multi-Criteria Decision Analysis (MCDA) Ranking Architecture

Rather than relying solely on raw follower counts (which penalizes high-performing micro-creators) or raw engagement rates (which penalizes macro-creators due to platform power laws), the system implements an explainable Multi-Criteria Decision Analysis (MCDA) framework (`src/influencer_ranking.py`).

`Influence Score = 0.40 × Engagement + 0.25 × Reach + 0.20 × Cadence + 0.15 × Content`

Each scoring dimension is computed on a min-max normalized scale \([0, 100]\):

- **Engagement Score (40% Weight):** Evaluates log-scaled engagement rate and interaction efficiency relative to peer benchmarks.
- **Audience Reach Score (25% Weight):** Evaluates log-scaled verified follower count strictly within the [0, 1,000,000) boundary.
- **Posting Consistency Score (20% Weight):** Evaluates cadence regularity and monthly posting volume.
- **Content Score (15% Weight):** Evaluates hashtag diversity and thematic specialization depth.

Deterministic Tie-Breaking Logic: Ties in final Influence Score are broken sequentially and deterministically by: `Engagement Score → Reach Score → Consistency Score → Content Score → Total Followers → Username (A—Z)`. This ensures 100% reproducible rank order across all 2,561 creators without arbitrary random outcomes.

5. Top 10 Influencers Overall

The table below highlights the top 10 ranked creators across the entire dataset according to the composite MCDA scoring model:

#1
@d.d.on.the.spot
Entertainment
127,301
69.82%
30.0/mo
88.10
Balanced High Reach & ER
#2
@sprite_india
Other
134,429
86.32%
30.0/mo
86.46
Balanced High Reach & ER
#3
@adorableruffs
Pets & Animals
144,794
11.16%
30.0/mo
81.00
Balanced High Reach & ER
#4
@robertmartinezreels
Parenting
249,663
19.38%
30.0/mo
79.13
Balanced High Reach & ER
#5
@loveandlight707
Other
4,467
31.23%
30.0/mo
78.48
Exceptional Community ER
#6
@jurika.juju
Pets & Animals
14,825
1672.31%
17.3/mo
78.05
Exceptional Community ER
#7
@tchalametcutie
Other
4,293
23.90%
30.0/mo
77.60

Exceptional Community ER
#8
@cryptoland_en
Technology
108,405
27.64%
16.9/mo
76.43
Balanced High Reach & ER
#9
@bhagyalakshmi_offical_9398
Other
2,962
37.91%
15.4/mo
75.34
Exceptional Community ER
#10
@masculineehub
Education
5,178
62.34%
30.0/mo
74.99
Exceptional Community ER

6. Cross-Category Landscape & Benchmark Comparisons

Creator dynamics differ fundamentally across industry niches. For example, niche technical communities often record higher median engagement rates than saturated lifestyle feeds, while commercial lifestyle niches command larger median audiences. The table below presents the verified benchmarks and top-ranked creator across all 19 niches:

Technology
1,542
515
6.81%
48.21
@cryptoland_en
#8
76.43
Other
473
8,611
1.18%
47.02
@sprite_india
#2
86.46
Business & Finance

113
470
5.28%
48.10
@rodezweb
#82
64.51
Fashion & Beauty
85
1,145
1.72%
49.23
@__makeuptipsbywanda__
#30
68.33
Art & Design
71
584
7.17%
51.67
@im.medusax
#13
73.31
Education
47
682
7.83%
49.20
@masculineehub
#10
74.99
Travel
36
1,428
1.82%
48.06
@ingenierosyviajeros
#96
63.82
Lifestyle
35
3,524
2.00%
51.43
@nilfkifa
#36
67.25
Fitness & Health
29
2,383
1.18%
49.54
@sosofitelhuahin
#18
71.00
Food & Cooking
28
10,510
0.76%
48.66
@bayareafilipinos

#292
58.45
Photography
28
1,046
4.40%
50.51
@cristianschrik
#27
68.50
Entertainment
17
36,259
0.79%
53.29
@d.d.on.the.spot
#1
88.10
Parenting
15
17,168
0.37%
47.27
@robertmartinezreels
#4
79.13
Pets & Animals
12
1,624
6.76%
58.15
@adorableruffs
#3
81.00
Sports
10
1,132
3.82%
45.52
@eagleswwtfc
#597
54.20
Gaming
8
2,848
2.74%
46.47
@draft5gg
#12
73.77
Music
6
5,274
2.18%
47.66
@gonzalolagioia
#444
56.15
Automotive
3
10,230

5.29%
50.57
@the.__.hyper.__.boy
#78
64.66
Home & Garden
3
16,604
0.04%
46.10
@chloedomecq
#578
54.47

7. Unsupervised Creator Segmentation (K-Means Clustering)

To discover natural creator segments without subjective human bias, K-Means clustering was executed on 64 standardized features across *k = 2 … 10*. Evaluation metrics confirmed **k = 3** as the optimal clustering configuration:

- **Peak Multi-Segment Silhouette Score:** 0.3318 (superior cluster cohesion and separation).
- **Lowest Davies-Bouldin Index:** 1.2996 (minimal intra-cluster scatter relative to inter-cluster separation).
- **Elbow Criterion:** Inertia decreased smoothly from 138,590 (*k = 2*) to 120,442 (*k = 3*), showing diminishing returns beyond *k = 3*.

Cluster 0: Active Macro Tier
472
18.43%
11,241
2.59%
17.3 posts/mo
Brand awareness campaigns; broad sponsor visibility
Cluster 1: Emerging High-ER Nano
1,859
72.59%
430
6.83%
5.7 posts/mo
Niche community advocacy; high conversion efficiency
Cluster 2: High-Reach Low-ER
230
8.98%
23,271
0.21%
12.3 posts/mo
Dormant or saturated accounts; secondary review required

8. Statistical Anomaly & Metric Quality Audit

The anomaly audit combines high-dimensional Isolation Forest modeling (`contamination=0.05`, 100 estimators, `random_state=42`) with rule-based edge case detection to identify statistically unusual profiles while preserving data integrity:

- **Isolation Forest Multivariate Outliers:** Identified 128 profiles (5.00% of dataset) exhibiting unusual multivariate combinations of reach, interaction volume, and posting cadence.
- **Extreme Engagement Rates (> 100%):** Identified 9 profiles. These represent mathematically valid social media dynamics where posts reach Instagram's Explore page or Reels recommendations, accumulating non-follower interactions on smaller follower bases.
- **Hidden Like Counts:** Detected 46 profiles where likes equal 0 while active comments are positive. This directly reflects Instagram's native UI privacy feature allowing creators to hide public like counts, rather than automated engagement failure.
- **Dormant Audiences:** Identified 62 profiles with followings > 10,000 recording zero likes, signaling inactive or disconnected historical audiences.

9. Heuristic Automation Risk Screening

The Automation Risk Index (`outputs/automation_insights.csv`) provides an exploratory screening metric [0, 100] to guide sponsor due diligence. It synthesizes four behavioral indicators:

1. **Cadence Risk (25% Weight):** Flags rigid daily syndication (≥ 28 posts/month).
2. **Engagement Disconnect Risk (30% Weight):** Flags large audience followings paired with suppressed ER (< 2.0%).
3. **Interaction Asymmetry Risk (25% Weight):** Flags zero-like anomalies or severe like-to-comment imbalances.
4. **Metadata Absence Risk (20% Weight):** Flags missing hashtags and themes indicating uncurated automated feeds.

Low Risk
< 25.0
1,958
76.45%
Organic metrics confirmed; approved for standard engagement.
Moderate Risk
25.0 – 50.0
347
13.55%
Typical creator metrics; standard contract vetting.
High Risk
50.0 – 75.0
193
7.54%
Secondary review recommended; inspect comment quality.
Elevated Audit Required
≥ 75.0

63
2.46%
Detailed pre-contract audit required; verify native Insights.

10. Methodological Limitations & Ethical Safeguards

Critical Ethical Notice:
1. **Statistical Outlier ≠ Fraud:** An anomaly indicates an unusual statistical pattern in high-dimensional feature space. It does NOT establish artificial bots or fraudulent manipulation. Legitimate viral spikes routinely generate high engagement rates.
2. **Heuristic Score ≠ Calibrated Probability:** The Automation Risk Index is an empirical heuristic screening score, not a calibrated Bayesian probability. Neutral terminology such as *"Review Recommended"* is enforced to prevent unjustified commercial harm.
3. **Absence of Ground Truth:** Public Instagram metadata does not contain verified ground-truth bot labels. Supervised ML was deliberately avoided to prevent training on unverified proxy targets.
4. **Platform Compliance:** In accordance with Instagram Terms of Service, live authenticated scraping and credential stuffing were strictly avoided. All data was ethically sourced from public research archives.

11. Production Dashboard & Repository Integration

All analytical findings, interactive filters, leaderboards, and segment benchmarks are publicly accessible in the deployed production application:

- **Deployed Interactive Streamlit Dashboard:** https://k2koturvzr7f2jfxrf4gtz.streamlit.app/
- **Private GitHub Project Repository:** https://github.com/BrewAndCode/-instagram-influencer-analytics

The application enables commercial marketing managers to filter creators by category, cluster segment, follower tier, and keyword search, inspect decomposed MCDA dimension scores, view interactive Plotly visualizations, and conduct sponsor due diligence in real time.

12. Deliverable Verification & Sign-Off

All Phase 2 deliverables for the SR NEXT Internship Program have been verified:

- **Ranked Influencer Deliverable:** `outputs/ranked_influencers.csv` — Exactly 2,561 records, 16 columns, zero nulls, strictly monotonic ranking order.
- **Phase 2 Analysis Report:** `reports/Phase_2_Analysis_Report.pdf` — Verified multi-page submission PDF.
- **Phase 1 Baseline Dataset:** `data/processed/instagram_influencers_final_clean.csv` — 100% unmodified (MD5: `e1b0b2b7687ab4fe4956cabcc808b8b4`).

— END OF PHASE 2 ANALYSIS REPORT —